

BAHARIA, Egypt

WMO: 624200

Lat: 28.333N

Lon: 28.867E

Elev: 427

StdP: 14.47

Time Zone: 2.00 (E02)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.9	43.0	19.0	14.5	67.6	22.6	17.2	65.7	16.0	61.4	14.3	62.3	1.1	200	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.9	107.6	68.7	105.1	68.5	102.7	68.5	73.3	92.3	72.3	92.3	71.5	92.2	7.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.1	108.7	78.5	67.4	102.2	78.7	65.7	96.4	78.6	37.1	91.6	36.2	92.0	35.5	91.6	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	13.2	11.7	37.8	112.0	2.8	3.4	35.8	114.4	34.2	116.4	32.6	118.3	30.5	120.8
			32.8	75.4	2.6	1.3	30.9	76.4	29.5	77.1	28.0	77.9	26.2	78.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	73.8	56.8	60.0	66.4	73.5	80.3	85.4	87.6	88.1	83.9	76.6	67.4	59.4	(6)
		DBStd	12.10	3.95	5.62	6.46	6.94	5.82	4.59	3.38	3.00	4.10	4.86	4.74	4.23	(7)
		HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
		HDD65	684	256	161	56	5	0	0	0	0	0	0	25	181	(9)
		CDD50	8705	212	280	507	704	940	1062	1165	1182	1016	823	521	293	(10)
		CDD65	3913	2	20	99	259	475	612	700	717	566	358	96	9	(11)
		CDH74	52727	54	268	1268	3234	6431	8945	10508	10558	7168	3426	770	97	(12)
(13)		CDH80	30961	7	87	525	1644	3754	5656	6726	6645	4145	1551	202	19	(13)
(14)	Wind	WSAvg	4.6	3.8	4.3	5.0	5.4	5.3	5.1	4.4	4.8	4.3	3.4	3.5		(14)
(15)	Precipitation	PrecAvg	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(15)
(16)		PrecMax	0.6	0.2	0.3	0.3	0.5	0.3	0.0	0.0	0.0	0.0	0.1	0.2	0.1	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.2	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	79.8	88.9	98.3	104.5	110.7	111.9	110.7	109.3	105.9	100.5	91.2	82.6		(19)
(20)		0.4% MCWB	56.2	59.8	62.6	65.0	66.7	67.4	69.4	70.0	69.2	68.5	64.7	60.9		(20)
(21)		2% DB	74.2	81.6	91.3	99.2	104.3	107.0	106.2	105.8	102.7	95.5	85.0	76.0		(21)
(22)		2% MCWB	54.6	56.1	60.1	63.5	66.1	68.2	69.3	69.5	68.9	67.2	62.9	57.4		(22)
(23)		5% DB	70.6	76.1	85.6	94.5	99.9	103.4	103.6	103.3	99.6	91.8	81.3	72.2		(23)
(24)		5% MCWB	53.5	54.7	58.4	61.8	65.0	67.6	69.1	69.8	68.8	66.3	61.7	56.5		(24)
(25)		10% DB	67.4	72.4	80.6	89.5	96.0	100.0	101.0	100.8	96.5	88.2	78.2	69.5		(25)
(26)		10% MCWB	52.8	53.8	56.8	60.7	64.2	67.4	69.0	69.7	68.4	65.6	60.6	55.5		(26)
(27)		0.4% WB	59.0	61.2	65.4	67.7	69.3	72.2	73.8	75.2	73.6	72.0	67.5	62.1		(27)
(28)		0.4% MCDB	74.8	84.1	89.0	96.7	100.1	98.1	94.9	90.7	90.0	92.1	82.9	77.5		(28)
(29)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	56.5	58.0	61.7	65.1	67.7	70.9	72.6	73.7	72.0	69.8	65.0	59.4		(29)
(30)		2% MCDB	69.8	76.2	86.5	94.4	96.7	97.6	94.3	91.3	89.6	87.2	80.3	72.2		(30)
(31)		5% WB	54.9	56.1	59.5	63.1	66.6	69.8	71.7	72.7	71.1	68.3	63.4	57.7		(31)
(32)		5% MCDB	68.0	72.7	81.5	90.4	94.5	95.2	93.5	91.5	89.3	84.8	77.6	70.1		(32)
(33)		10% WB	53.5	54.6	57.8	61.2	65.4	68.6	70.8	71.8	70.2	67.1	61.8	56.1		(33)
(34)		10% MCDB	66.1	70.5	77.8	86.2	91.9	93.6	92.8	91.6	89.1	83.0	75.3	68.1		(34)
(35)	Mean Daily Temperature Range	MDBR	22.6	23.7	26.1	27.1	28.1	28.3	27.2	25.9	25.2	23.5	22.8	22.1		(35)
(36)		5% DB MCDBR	27.4	29.3	31.9	33.1	33.2	32.2	30.2	28.9	29.3	28.0	26.1	25.2		(36)
(37)		5% DB MCWBR	14.0	13.0	12.4	11.5	10.3	8.5	8.4	8.2	8.3	9.2	10.8	12.5		(37)
(38)		5% WB MCDBR	23.7	25.5	28.7	29.4	28.5	28.1	26.7	25.4	25.2	23.9	23.6	23.1		(38)
(39)		5% WB MCWBR	13.1	11.9	11.9	10.8	9.7	8.0	7.6	7.3	7.9	8.6	10.1	12.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.393	0.435	0.481	0.556	0.562	0.488	0.466	0.460	0.467	0.454	0.404	0.389		(40)
(41)		taud	2.036	1.907	1.789	1.649	1.664	1.849	1.922	1.947	1.920	1.940	2.046	2.061		(41)
(42)		Ebn at Noon	261	260	256	242	240	257	263	264	257	251	255	256		(42)
(43)		Edh at Noon	46	57	68	81	80	66	61	59	59	54	45	43		(43)
(44)	All-Sky Solar Radiation	RadAvg	1275	1560	1950	2233	2438	2584	2527	2362	2093	1708	1364	1188		(44)
(45)		RadStd	28	67	52	54	54	35	20	26	39	27	31	27		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	+1.03	+1.35	N/A	N/A	N/A	+1.21	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page